Boston House Price Prediction (Low Code)

OLS Multiple Linear Regression Modeling of Boston Housing Prices Using the Boston SMSA 1970 Dataset

Methodology and Workflow

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1. Executive Summary

The dataset analyzed is the **Boston Housing dataset**, collected in 1970 from the **Boston Standard Metropolitan Statistical Area (SMSA)**. The attributes (independent variables, features, or predictors) are evaluated as essential predictors of home prices in Boston suburbs and towns. A **multiple linear regression model** is built to effectively capture variation in Boston home prices across neighborhoods and vicinities. The analysis comprises exploratory data analysis, feature engineering, and thorough diagnostics for multicollinearity to build a robust, accountable **ordinary least squares (OLS)** multiple linear regression model.

1.1 Key EDA Findings

- **LSTAT** (percentage of lower-status population), **RM** (average number of rooms per dwelling), and **DIS** (weighted distances to five Boston employment centers) revealed strong relationships with home values.
- Houses farther from Boston employment centers were more likely to have large-lot zoning (**ZN**).

- Houses in areas with higher index of accessibility to radial highways (**RAD**) usually had higher property-tax rates (**TAX**).
- Outlier values in each independent numerical variable were retained because they reflect real-world conditions.

1.2 Preprocessing Highlights

- **Log transformation** applied to the target **MEDV** variable to address skewness and improve linearity.
- **RAD** was one-hot encoded to avoid absolute multicollinearity; **CHAS** remained binary.
- **VIF analysis** conducted to identify and remove superfluous dummy variables with infinite or undefined VIF values.

1.3 Final Model Results

The final OLS multiple linear regression model has an **R-squared of approximately 0.70**, explaining ~70% of the variation in home prices. **RMSE**, **MAE**, and **MAPE** are small and stable; no evidence of overfitting between training and test datasets. The most influential predictors (key drivers) are:

- **CRIM** — Per capita crime rate by town
- **CHAS** — Adjacency to the Charles River
- **NOX** — Nitric oxide concentration
- **RM** — Average number of rooms per dwelling
- **DIS** — Weighted distances to five Boston employment centers
- **PTRATIO** — Pupil-teacher ratio by town
- **LSTAT** — Percentage of lower status of the population

1.4 Coefficient Interpretations

Predictor	Direction	Interpretation
CRIM	Negative	Higher crime rates → lower home values
CHAS	Positive	Homes near the Charles River → valued more
NOX	Negative	High nitric oxide pollution → lower values
RM	Positive	More rooms → higher values
DIS	Negative	Farther from employment centers → lower values
PTRATIO	Negative	Higher pupil-teacher ratios → lower home values
LSTAT	Negative	Higher lower-status population percentage → lower values

The project provides data-driven discernment of the independent variables affecting Boston home pricing and delivers a statistically robust OLS model designed for effective interpretation and governance. Findings highlight the interdependence among environmental, zoning policy, transportation access, and neighborhood demographic factors that affect the Boston residential market in 1970. Real estate businesses, home buyers, property investors, and housebuilders can utilize the model for operational decisions, pricing strategies, investment dispositions, marketing emphasis, and risk evaluation.

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2. Business Problem Overview and Solution Approach

2.1 Business Problem Overview

Different organizations and stakeholders — home buyers, property investors, housebuilders — need to understand which factors determined house values in

Boston suburbs, towns, and neighborhoods in 1970. Without a data-driven approach, stakeholders risk incorrectly allocating resources, mispricing properties, and incorporating policies that do not capture achievable economic and community standards.

The core business problem is to correctly categorize the **key performance variables** that deterministically determine housing prices in Boston, enabling stakeholders to make informed decisions about development, investment, and policy design.

2.2 Analysis Explanation in Achieving Business Objective

The business objective is to accurately predict housing prices in Boston across different towns and suburbs using compiled locality features. To meet this objective, it is critical to identify the key factors driving house prices. The analysis transforms housing and neighborhood data into legitimate insights by applying EDA methods, data preprocessing techniques, and framing an OLS multiple linear regression model.

The project assesses how each factor affects the target **MEDV** variable. The analysis ensures the OLS model is interpretable and valid after addressing skewed distributions, categorical encoding, and multicollinearity.

2.3 Solution Approach / Methodology

The project follows a data-driven, multi-faceted methodology using **Python-based statistical modeling** and computations. The approach emphasizes Python-powered statistical analyses (code snippets provided in the appendix). Key methodology steps include:

- Descriptive analytics and correlation analysis
- Data cleaning
- Log-transforming the target MEDV variable to improve linearity
- One-hot encoding the RAD independent variable
- VIF diagnostics to remove redundant categorical dummy variables

- Train-test split to evaluate model performance
- Removing non-statistically significant independent variables
- Cross-validation for model validation

The methodology yields a stable, high-quality model that best explains variation in housing prices while providing infrastructure for decision-making in real estate planning, pricing, and policy development.

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3. Data Overview

3.1 Dataset Description

The dataset is drawn from the **Boston Standard Metropolitan Statistical Area (SMSA)** in 1970. It includes **506 unique homes** (each represented by a row) and **13 columns**. No missing values and no duplicate entries were found.

3.2 Categorical Variables

Two categorical independent variables are present in the dataset:

- **CHAS** (Charles River dummy variable): Binary (1 = river adjacency, 0 = none).
- **RAD** (Index of accessibility to radial highways): Numeric codes representing distinct categories (values: 1, 2, 3, 4, 5, 6, 7, 8, and 24).

3.3 Numerical Variables

Ten continuous numerical independent variables:

Variable	Description
CRIM	Per capita crime rate by town
ZN	Proportion of residential land zoned for lots over 25,000 sq ft

Variable	Description
INDUS	Proportion of non-retail business acres per town
NOX	Nitric oxide concentration (parts per 10 million)
RM	Average number of rooms per dwelling
AGE	Proportion of owner-occupied units built before 1940
DIS	Weighted distances to five Boston employment centers
TAX	Full-value property-tax rate per $10,000
PTRATIO	Pupil-teacher ratio by town
LSTAT	Percentage of lower status of the population

Target variable:

- **MEDV** — Median value of owner-occupied homes in $1,000

3.4 Summary Descriptive Statistics

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
CRIM	506.0	3.61	8.60	0.01	0.08	0.26	3.68	88.98
ZN	506.0	11.36	23.32	0.00	0.00	0.00	12.50	100.00
INDUS	506.0	11.14	6.86	0.46	5.19	9.69	18.10	27.74
CHAS	506.0	0.07	0.25	0.00	0.00	0.00	0.00	1.00
NOX	506.0	0.55	0.12	0.38	0.45	0.54	0.62	0.87
RM	506.0	6.28	0.70	3.56	5.89	6.21	6.62	8.78
AGE	506.0	68.57	28.15	2.90	45.02	77.50	94.07	100.00
DIS	506.0	3.80	2.11	1.13	2.10	3.21	5.19	12.13
RAD	506.0	9.55	8.71	1.00	4.00	5.00	24.00	24.00
TAX	506.0	408.24	168.54	187.00	279.00	330.00	666.00	711.00

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
PTRATIO	506.0	18.46	2.16	12.60	17.40	19.05	20.20	22.00
LSTAT	506.0	12.65	7.14	1.73	6.95	11.36	16.96	37.97
MEDV	506.0	22.53	9.20	5.00	17.02	21.20	25.00	50.00

3.5 Key Descriptive Observations

- **CRIM:** Average 3.61, heavily right-skewed; most towns have low crime, a few have extremely high rates up to 88.98.
- **ZN:** Majority of towns have zero large-lot zoning; median is 0; outliers range 30–100.
- **INDUS:** Moderately right-skewed; median 9.69; suggests moderate industrialization across the SMSA.
- **CHAS:** Only 6.9% of homes are river-adjacent (mean 0.07); uncommon environmental amenity.
- **NOX:** Moderate pollution levels; median 0.54; mix of cleaner suburban and more polluted urban areas.
- **RM:** Nearly normal distribution; median 6.21; most homes are moderate-sized (5–7 rooms).
- **AGE:** Most homes are older (median 77.50); slight bimodality between newer and very old homes.
- **DIS:** Right-skewed; most homes are closer to employment centers (median 3.21).
- **TAX:** Moderately right-skewed; median 330 per $10,000; significant burden range across towns.
- **PTRATIO:** Clustered distribution; median 19.05; most towns have similar pupil-teacher ratios.

- **LSTAT:** Right-skewed; median 11.36; most towns have lower proportions of lower-status residents.
- **MEDV:** Right-skewed; median $21,200; strong concentration $15K–$25K; outliers up to $50K.

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4. Exploratory Data Analysis (EDA) and Data Preprocessing

4.1 Univariate Analysis

The univariate analysis methodology employed the following visualization techniques:

- **Bar plots** were used for categorical variables (CHAS, RAD).
- **Combined histogram and boxplot** functions were used for all numerical variables.
- Median shown as a solid black line in histograms; mean shown as a triangle in boxplots and a dashed line in histograms.

CHAS — Charles River Dummy Variable (Graph 1)

93.1% of homes are not river-adjacent; only 6.9% are located near the Charles River. This heavily imbalanced distribution indicates that river adjacency is a rare environmental amenity. River-adjacent homes may correlate with higher MEDV values due to the desirability of waterfront proximity.

RAD — Index of Accessibility to Radial Highways (Graph 2)

The distribution of RAD is uneven. RAD=24 accounts for 26.1% of all observations, RAD=5 for 22.7%, and RAD=4 for 21.7% — together comprising approximately 70% of all accessibility values. Higher RAD values could suggest lower MEDV due to increased noise and traffic exposure, higher NOX concentrations, and more commercial or industrial zoning in those areas.

CRIM — Per Capita Crime Rate (Graph 3)

CRIM is heavily right-skewed, with the vast majority of towns clustered near 0 and outliers ranging from approximately 15 to over 80. Higher crime rates are associated with lower MEDV, higher LSTAT, higher NOX, and denser urban areas. The extreme right tail reflects a small number of high-crime neighborhoods that substantially differ from the majority of the SMSA.

ZN — Proportion of Large-Lot Zoning (Graph 4)

ZN is heavily right-skewed with the majority of observations at 0, indicating that most towns lack large-lot residential zoning. Outliers range from 30 to 100, representing a few suburban towns with extensive large-lot zoning. These towns tend to have higher MEDV values, reflecting the premium on lower-density residential areas.

INDUS — Proportion of Non-Retail Business Acres (Graph 5)

INDUS is moderately right-skewed with a concentration in the 2–8 range. Higher INDUS values indicate greater industrial presence, which is generally associated with lower MEDV values due to environmental and noise externalities from commercial and manufacturing activities.

NOX — Nitric Oxide Concentration (Graph 6)

NOX is moderately right-skewed, with most towns falling in the 0.45–0.60 range. Lower NOX concentrations are associated with higher MEDV values and are characteristic of suburban, lower-density areas with less industrial activity and vehicular traffic.

RM — Average Number of Rooms (Graph 7)

RM follows a nearly normal distribution, with the most common range being 5–7 rooms per dwelling. A clear positive relationship exists: more rooms correspond to higher MEDV values. Outliers appear below 5 rooms and above 7 rooms, representing unusually small or large dwellings respectively.

AGE — Proportion of Pre-1940 Housing (Graph 8)

AGE exhibits slight bimodality, with the majority of homes having AGE values between 80 and 100, indicating that most owner-occupied units were built before 1940. Higher AGE values tend to correlate with lower MEDV, as older housing stock may require more maintenance and lack modern amenities.

DIS — Weighted Distance to Employment Centers (Graph 9)

DIS is significantly right-skewed, with most homes concentrated at distances of 1–4 units from employment centers. Proximity to employment is associated with higher MEDV values, reflecting the desirability of convenient commuting access to major job hubs in the Boston area.

TAX — Property Tax Rate (Graph 10)

TAX is moderately right-skewed, with a concentration at 250–400 per $10,000 of assessed value. Higher TAX values are generally associated with lower MEDV, as elevated tax burdens reduce the net value proposition for homeowners and investors.

PTRATIO — Pupil-Teacher Ratio (Graph 11)

PTRATIO displays a narrow, clustered distribution concentrated between 16 and 22. Higher pupil-teacher ratios are associated with lower MEDV values, reflecting the premium that homebuyers place on communities with better-funded schools and smaller class sizes.

LSTAT — Percentage of Lower-Status Population (Graph 12)

LSTAT is significantly right-skewed, with most observations concentrated in the 0–15% range. Higher LSTAT values are strongly correlated with lower MEDV values, making this one of the most influential predictors in the dataset.

MEDV — Median Home Value (Graph 13)

MEDV is right-skewed with a strong concentration between $15,000 and $25,000 (in 1970 dollars). Outliers appear in the $38,000–$50,000 range at the upper end and

below $5,000 at the lower end. The maximum value of $50,000 appears to be a censoring cap applied during data collection.

4.2 Bivariate Analysis — Correlation Analysis

A **Pearson correlation heatmap** was computed for all variables (Graph 14). The Pearson correlation coefficient (r) measures the strength and direction of the linear relationship between two continuous variables. A positive r indicates a positive linear relationship, a negative r indicates a negative linear relationship, and r near 0 indicates no linear relationship. Importantly, **correlation does not imply causation**.

Summary of 14 Most Significant Correlations

#	Variable Pair	r	Strength	Interpretation
1	RM vs. MEDV	0.70	Strong positive	More rooms → higher value
2	LSTAT vs. MEDV	−0.74	Strong negative	Higher lower-status % → lower value
3	PTRATIO vs. MEDV	−0.51	Moderate negative	Higher pupil-teacher ratio → lower value
4	NOX vs. MEDV	−0.43	Moderate negative	More pollution → lower value
5	DIS vs. MEDV	0.25	Weak positive	Farther from jobs → slightly higher value
6	CRIM vs. MEDV	−0.39	Moderate negative	More crime → lower value
7	INDUS vs. MEDV	−0.48	Moderate negative	More industrial → lower value
8	ZN vs. MEDV	0.36	Weak positive	More large-lot zoning → higher value
9	NOX vs. INDUS	0.76	Strong positive	Industrial areas →

#	Variable Pair	r	Strength	Interpretation
				higher pollution
10	DIS vs. NOX	−0.77	Strong negative	Farther from employment → less pollution
11	DIS vs. INDUS	−0.71	Strong negative	Suburban areas → less industrial
12	ZN vs. DIS	0.66	Moderate positive	Farther from centers → more large-lot zoning
13	ZN vs. AGE	−0.57	Moderate negative	Newer homes in low-density areas
14	RAD vs. TAX	0.91	Very strong positive	More highway access → higher tax rates (adjusted r = 0.25 after removing high-TAX outliers)

Scatterplot Analysis (Graphs 15–22)

Targeted scatterplots were generated for the most significant variable pairs to visually confirm the correlation patterns identified in the heatmap:

- **AGE vs. DIS (r = −0.75):** Inner-city suburbs closer to employment centers have a higher proportion of older homes. As distance increases, the housing stock becomes newer and more varied.
- **RAD vs. TAX (r = 0.91):** A strongly clustered relationship exists. Areas with RAD=24 consistently have the highest TAX values, creating two distinct clusters. The very strong correlation is driven primarily by this high-RAD/high-TAX cluster.

- **INDUS vs. TAX (r = 0.72):** More industrialized towns tend to have higher property-tax rates, reflecting the fiscal needs of municipalities with greater industrial infrastructure.
- **LSTAT vs. MEDV (r = −0.74):** House prices decrease as LSTAT increases, with a clear downward trend. MEDV appears capped at $50,000, creating a ceiling effect at the top of the distribution.
- **RM vs. MEDV (r = 0.70):** A significant upward slope is evident from 6 to 7.5 rooms. The relationship shows acceleration at higher room counts. MEDV is again capped at 50.
- **INDUS vs. NOX (r = 0.76):** Higher industrialization corresponds directly to higher pollution. Data points are tightly clustered at low INDUS/low NOX values and fan outward as industrialization increases.
- **AGE vs. NOX (r = 0.73):** Older homes are disproportionately located in denser, more polluted areas — consistent with historical urban development patterns.
- **DIS vs. NOX (r = −0.77):** Closer proximity to employment centers corresponds to higher NOX concentrations. Farther from employment centers, the air quality improves substantially, following a clear inverse relationship.

4.3 Data Preprocessing

4.3.1 Outlier Handling

All outlier values were **retained** because they are true, real, and meaningful, reflecting truthful socioeconomic characteristics of the Boston SMSA. Removing outliers would:

- Distort distributions and introduce artificial sanitization bias
- Reduce model transferability to new data
- Remove significant correlation patterns essential for accurate prediction

Validation Note

A separate OLS model was computed with capped outliers (CRIM, ZN, DIS, PTRATIO, LSTAT) and proved significantly weaker — yielding lower R-squared and higher RMSE, MAE, and MAPE values. This confirms the decision to retain all outliers.

Important: Skewed independent variables (CRIM, INDUS, DIS, LSTAT) were kept on their original scale. No log transformation was applied to independent variables because coefficient interpretation would become substantially more complex without proportional interpretive benefit.

4.3.2 Log Transformation of MEDV

A **log transformation** was applied to the target MEDV variable for the following reasons:

- Before transformation, MEDV had unstable variance and showed no clear linear relationships with independent variables.
- The log transformation pulls in the right tail, making the distribution more symmetric.
- Residuals become normally distributed and exhibit constant variance.
- Independent variables add fixed amounts to the log-transformed target, enabling percentage-based interpretation.
- This leads to improved model performance and more trustworthy regression assumptions.

After transformation, **MEDV_log** appears nearly normally distributed with little skew (Graph 23).

4.3.3 Categorical Variable Encoding

- **CHAS:** Already in preferred binary structure (0/1) — no encoding needed.
- **RAD:** One-hot encoded using **pd.get_dummies** with **drop_first=True** to prevent the model from assuming meaningful distance between categories,

ordinal meaning, or equal spacing. This approach removes spurious numeric relationships, enables independent effects for each group, and improves interpretability.

4.3.4 Data Partitioning

- **Features (X):** All independent variables with intercept term added via **sm.add_constant**.
- **Target (Y):** MEDV_log.
- **Train-test split:** 80% training (404 rows, 20 columns) / 20% testing (102 rows, 20 columns).
- An 80-to-20 split was chosen given the small dataset size (506 observations) to balance training sufficiency with testing reliability.

4.3.5 Multicollinearity Diagnostics (VIF Analysis)

Multicollinearity occurs when two or more independent variables are highly correlated, carrying the same information. Its consequences include:

- Unstable coefficients that fluctuate with minor data changes
- Inflated variance of coefficient estimates
- Inaccurate statistical significance results
- Difficulty interpreting individual coefficient effects
- Numerical computation issues

The **Variance Inflation Factor (VIF)** function was called iteratively to identify and remove multicollinear features:

First VIF Check (Table 6): RAD_24 had VIF = 15.63 (exceeding the threshold of 10) — dropped from the model.

Second VIF Check (Table 7): TAX had VIF = 6.06 (exceeding the threshold of 5) — dropped from the model.

Final VIF Check (Table 8): All remaining features have VIF below 5, confirming that multicollinearity has been adequately addressed.

Feature	VIF
const	620.12
CRIM	1.64
ZN	2.61
INDUS	3.48
CHAS	1.11
NOX	4.52
RM	1.96
AGE	3.31
DIS	4.42
PTRATIO	2.20
LSTAT	3.03
RAD_2	1.52
RAD_3	1.78
RAD_4	2.14
RAD_5	2.75
RAD_6	1.46
RAD_7	1.38
RAD_8	1.51

Boolean RAD dummy variables were converted to integer type before model fitting to ensure compatibility with the OLS estimation routine.

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5. Linear Regression and Its Assumptions

5.1 Evaluation Metrics Definitions

Metric	Definition
Residuals	The differences between the actual observed values and the predicted values produced by the model. Residuals indicate how far off each prediction is from reality.
R-squared (R^2)	The proportion of variance in the dependent variable explained by the independent variables. Ranges from 0 to 1; closer to 1 indicates a better fit.
Adjusted R-squared	A modified version of R-squared that penalizes for the number of predictors in the model. Corrects for R-squared inflation caused by adding variables that do not genuinely improve the model.
p-value	The probability that the observed relationship between a variable and the target is due to random chance. A p-value below 0.05 indicates statistical significance; above 0.05 indicates the relationship is not statistically significant.
RMSE	Root Mean Squared Error — the typical size of prediction errors, in the same units as the target. Computed by squaring residuals, averaging them, and taking the square root.
MAE	Mean Absolute Error — the average size of prediction errors using absolute residuals. Treats every error equally regardless of direction.
MAPE	Mean Absolute Percentage Error — the average percentage difference between actual and predicted values. Converts errors to percentages for scale-independent interpretation.

Performance Goal

For optimal model performance, RMSE, MAE, and MAPE should be as low as possible, while R-squared should be as close to 1 as possible.

5.2 Model 1 (model1) — Full Feature OLS Model

Model 1 Summary (Table 11):

Parameter	Value
Dependent Variable	MEDV_log
Method	Least Squares (OLS)
Observations	404
Df Residuals	386
Df Model	17
R-squared	0.780
Adjusted R-squared	0.770
F-statistic	80.57
Prob (F-statistic)	1.98e-115
Durbin-Watson	1.948

Model 1 Coefficient Table:

Variable	Coef	Std Err	t	P>|t|
const	4.2989	0.240	17.893	0.000
CRIM	−0.0098	0.001	−7.238	0.000
ZN	0.0013	0.001	1.953	0.052
INDUS	0.0002	0.003	0.063	0.950
CHAS	0.1049	0.038	2.790	0.006
NOX	−0.8391	0.176	−4.779	0.000
RM	0.0718	0.020	3.663	0.000
AGE	0.0004	0.001	0.651	0.516
DIS	−0.0523	0.010	−5.461	0.000

Variable	Coef	Std Err	t	P>|t|
PTRATIO	−0.0365	0.007	−5.521	0.000
LSTAT	−0.0297	0.002	−12.750	0.000
RAD_2	−0.0126	0.052	−0.240	0.810
RAD_3	0.0573	0.048	1.202	0.230
RAD_4	−0.0466	0.035	−1.348	0.178
RAD_5	−0.0225	0.038	−0.589	0.556
RAD_6	−0.0874	0.053	−1.664	0.097
RAD_7	0.0665	0.060	1.108	0.269
RAD_8	0.0107	0.057	0.186	0.852

Statistically significant variables ($p < 0.05$): CRIM, CHAS, NOX, RM, DIS, PTRATIO, LSTAT.

Non-significant variables ($p > 0.05$): ZN (0.052), INDUS (0.950), AGE (0.516), RAD_2 (0.810), RAD_3 (0.230), RAD_4 (0.178), RAD_5 (0.556), RAD_6 (0.097), RAD_7 (0.269), RAD_8 (0.852).

Model 1 Performance (Table 12):

Data	RMSE	MAE	MAPE	R^2
Train	0.1896	0.1344	4.6719	0.7182
Test	0.2055	0.1553	5.4412	0.6154

Observations: Model 1 generalizes fairly well with no evidence of overfitting. MAPE of approximately 5% on the test set indicates strong predictive accuracy. Mild tail errors exist but overall model stability is confirmed by the consistency between training and test metrics.

5.3 Model 2 (model2) — Refined OLS Model

After removing non-significant variables (ZN, INDUS, AGE, RAD_2 through RAD_8), the refined model retains only the **seven statistically significant predictors**.

Model 2 Summary (Table 13):

Parameter	Value
R-squared	0.771
Adjusted R-squared	0.767
F-statistic	190.4
Prob (F-statistic)	1.60e-122
Df Model	7
Durbin-Watson	1.988

Model 2 Coefficient Table:

Variable	Coef	Std Err	t	P>|t|
const	4.1437	0.205	20.221	0.000
CRIM	−0.0087	0.001	−7.036	0.000
CHAS	0.1070	0.037	2.897	0.004
NOX	−0.7785	0.143	−5.463	0.000
RM	0.0911	0.018	4.929	0.000
DIS	−0.0418	0.007	−5.748	0.000
PTRATIO	−0.0376	0.005	−7.413	0.000
LSTAT	−0.0290	0.002	−13.535	0.000

All predictors are statistically significant ($p < 0.05$).

Model 2 Performance (Table 14):

Data	RMSE	MAE	MAPE	R^2
Train	0.1935	0.1380	4.7822	0.7029
Test	0.2058	0.1574	5.5155	0.6164

Model Selection Rationale

Model 2 was chosen as the preferred model for the following reasons:

- Slightly higher R-squared on the test data compared to Model 1
- No statistically weak variables diluting the model
- No distortion of signs and magnitudes in coefficients
- Strengthened, stabilized, and recovered effects for all significant variables
- Higher overall interpretability, stability, and alignment with economic intuition

5.4 Model 2 Regression Equation

The final OLS regression equation for the preferred model is:

$$\log(MEDV) = 4.1437 - 0.0087 \times CRIM + 0.1070 \times CHAS - 0.7785 \times NOX + 0.0911 \times RM - 0.0418 \times DIS - 0.0376 \times PTRATIO - 0.0290 \times LSTAT$$

5.5 Coefficient Interpretation

Because the target variable is log-transformed, each coefficient represents an approximate **percentage change** in predicted MEDV for a one-unit increase in the corresponding predictor, holding all other variables constant:

Variable	Coefficient	Interpretation
CRIM	−0.0087	A one-unit increase in per capita crime rate is associated with an approximately **0.87% decrease** in predicted home price.
CHAS	0.1070	Homes adjacent to the Charles River command

Variable	Coefficient	Interpretation
		an approximately **10.7% premium** over non-river-adjacent homes.
NOX	−0.7785	A one-unit increase in nitric oxide concentration is associated with an approximately **7.8% decrease** in predicted home price per 0.1 unit increase — the strongest negative predictor.
RM	0.0911	Each additional room per dwelling is associated with an approximately **9.1% increase** in predicted home price.
DIS	−0.0418	Each additional unit of weighted distance from employment centers is associated with an approximately **4.2% decrease** in predicted home price.
PTRATIO	−0.0376	A one-unit increase in the pupil-teacher ratio is associated with an approximately **3.8% decrease** in predicted home price.
LSTAT	−0.0290	A one-percentage-point increase in lower-status population is associated with an approximately **2.9% decrease** in predicted home price.

Predictor Influence Ranking:

- **Strongest predictors:** NOX, RM, LSTAT
- **Moderate predictors:** PTRATIO, DIS, CHAS
- **Weakest predictor:** CRIM

5.6 Assumption Checks for Model 2

Assumption 1 — Mean of Residuals = 0

Result: −4.81e−15 (effectively zero).

Conclusion: Assumption met. There is no systematic over-prediction or under-prediction. The model is unbiased on average.

Assumption 2 — Homoscedasticity (Goldfeld-Quandt Test)

Test Statistic	Value
F-statistic	1.0881
p-value	0.2786

Since $p > 0.05$, we fail to reject the null hypothesis. Residual variances are similar across the range of fitted values. There is **no statistical evidence of heteroscedasticity**. The assumption is satisfied.

Assumption 3 — Linearity of Variables (Graph 24)

The residual plot (residuals vs. fitted values) shows **no noticeable strong pattern**. Residuals are randomly and uniformly scattered along the x-axis with no systematic curvature, funneling, or clustering. The linearity condition is satisfied. Constant variance is confirmed, and standard errors and p-values are trustworthy.

Assumption 4 — Normality of Error Terms

Q-Q Plot (Graph 25): Points stay close to the theoretical red line throughout most of the distribution, with only minor deviations at the tails. The residuals follow a normal distribution.

Residual Histogram (Graph 26): The histogram displays a bell-shaped curve peaking near zero, symmetric on both sides. There is no evidence of asymmetry or extreme outliers. The histogram findings are fully aligned with the Q-Q plot, confirming the normality assumption.

5.7 Cross-Validation

A **10-fold cross-validation** technique was applied to:

- Prevent overfitting by evaluating performance on unseen data folds
- Provide reliable, averaged performance estimates
- Corroborate that model complexity is appropriate
- Detect potential data issues or instabilities
- Use data more effectively by rotating training and testing partitions

Cross-Validation Results:

Metric	Mean	Std (±)
R-squared	0.729	0.239
Mean Squared Error	0.040	0.023

The model explains approximately **73% of variance** across all folds. The small average squared error (0.040) is aligned with the strong R-squared. The variability indicates some folds perform consistently well; MSE variability shows moderate accuracy across subsets, which is expected given the small dataset size.

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6. Conclusion

6.1 Actionable Insights

1. **Improve housing quality:** Allocate resources to renovations and building larger, multi-room houses. RM is a strong positive predictor of home value.
2. **Support environmental and pollution-reduction efforts:** Identify high-NOX neighborhoods and implement eco-friendly buffers such as tree planting, green corridors, and traffic rerouting to reduce nitric oxide exposure.

3. **Enhance public transit connectivity:** Improve transportation links to employment hubs for distant neighborhoods to offset the negative DIS effect on home values.
4. **Invest in education:** Apportion funds to lower pupil-teacher ratios in underperforming schools. Improved PTRATIO directly correlates with increased home values.
5. **Promote community socioeconomic programs:** Develop initiatives that enhance socioeconomic conditions across towns and suburbs to reduce LSTAT and lift neighborhood home values.
6. **Implement crime-reduction measures:** Deploy crime hotspot identification, improve street lighting, and launch community safety initiatives to reduce CRIM and protect property values.
7. **Leverage river-adjacency amenity effects:** Construct recreational areas, parks, and waterfront amenities near bodies of water to capitalize on the CHAS premium effect.

6.2 Significance of Techniques Used

- Regression diagnostics and cross-validation ensure statistically sound, operationally reliable insights.
- Complex business dynamics are translated into clear, interpretable coefficient relationships.
- Residual analysis and assumption criteria validated the model's ability to generalize across datasets.
- Cross-validation corroborates forecasting stability and guards against overfitting.
- Feature significance testing identified the strongest-influence variables for targeted resource allocation.
- Multicollinearity checking via VIF ensured stable, interpretable coefficient estimates.

- Performance metrics (R-squared, RMSE, MAE, MAPE) provide full transparency into model durability and constraints.

6.3 Finalized Coefficient Interpretation

- Boston home values are predominantly determined by **structural features** (RM), **environmental quality** (NOX), **socioeconomic demographics** (LSTAT, PTRATIO), **home localities** (CHAS, DIS), and **crime rates** (CRIM).
- Buyers value clean environments, prominent community attributes, and superb-quality housing features.
- Crime plays a modest role compared to environmental and socioeconomic predictors in determining home values.
- By using the log transformation, stakeholders can apply percentage-based insights to quantify how cumulative changes in predictors translate into expected home prices.

6.4 Proposed Business Actions

#	Action Area	Description
1	**Pricing Strategy**	Set data-driven listing prices based on RM, LSTAT, NOX, and PTRATIO coefficients to accurately reflect neighborhood value drivers.
2	**Investment and Renovation**	Use CHAS and RM coefficients to prioritize features in renovation projects; calculate ROI for room additions and waterfront improvements.
3	**Site Selection**	Simulate future value scenarios under different zoning changes to optimize development location decisions.

#	Action Area	Description
4	**Policy Evaluation**	Evaluate the projected impact of interventions such as improving school funding (reducing PTRATIO) and lowering NOX levels on housing values.
5	**Customer Segmentation**	Group purchasers by preferences — room space (RM), school quality (PTRATIO), employment proximity (DIS) — to tailor offerings.
6	**Marketing**	Advertise properties emphasizing the strongest positive value drivers: Charles River adjacency (CHAS) and room count (RM).

6.5 Business Success Translation

8. **Minimized financial risk:** Quantitative evidence improves price prediction accuracy and reduces mispricing exposure.
9. **Targeted resource allocation:** Identifies the most critical predictors to avoid spending on factors with low impact.
10. **Competitive advantage:** Data-driven explanations provide differentiation over intuition-based market approaches.
11. **Strategic planning:** Enables long-term portfolio diversification and policy advocacy based on measurable coefficient effects.
12. **Stakeholder communication:** Clear, coefficient-based explanations enhance transparency and trust in decision-making.

6.6 Key Takeaways

- Increased home size (RM) and Charles River adjacency (CHAS) → **increased home values**.

- Nitric oxide pollution (NOX) → **strongly negates home values** (largest negative impact).
- Socioeconomic factors (LSTAT, PTRATIO) frame buyer demand — school quality and neighborhood demographics are significant determinants.
- Proximity to employment centers (DIS) matters — homes closer to jobs command higher prices.
- Lower crime rates (CRIM) → **increased home values**, though the effect is more modest relative to environmental and socioeconomic predictors.

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7. Workflow Summary

The following numbered workflow synthesizes the complete methodology applied throughout this project, from problem definition to stakeholder communication:

13. **Define the business problem and research objective** — predict Boston housing prices using neighborhood attributes to enable data-driven decision-making for stakeholders.
14. **Acquire the Boston Housing dataset** from the Boston SMSA (1970), containing 506 observations and 13 variables.
15. **Conduct data quality checks** — verify missing values, duplicates, and data types. Confirm dataset integrity (no missing values, no duplicates).
16. **Compute summary descriptive statistics** for all variables (count, mean, standard deviation, min, quartiles, max).
17. **Perform univariate analysis** using bar plots (categorical variables: CHAS, RAD) and combined histogram-boxplots (all numerical variables).
18. **Perform bivariate analysis** using a Pearson correlation heatmap and targeted scatterplots for significant pairs ($|r| \geq 0.7$).
19. **Handle outliers** — retain all outlier values as they reflect real-world socioeconomic conditions.

20. **Apply log transformation** to the target variable MEDV to improve distribution symmetry, stabilize variance, and satisfy regression assumptions.
21. **One-hot encode the RAD categorical variable** using drop_first=True; retain CHAS as binary (no encoding needed).
22. **Partition the dataset** into features (X) and target (Y); add intercept term via sm.add_constant.
23. **Split data** into 80% training (404 observations) and 20% testing (102 observations) sets.
24. **Conduct iterative VIF analysis** to remove multicollinear variables: RAD_24 (VIF = 15.63) and TAX (VIF = 6.06).
25. **Convert boolean dummy variables** to integer type for OLS compatibility.
26. **Fit Model 1** (full feature OLS model) and evaluate p-values for all predictors.
27. **Remove non-significant variables** ($p > 0.05$): ZN, INDUS, AGE, RAD_2 through RAD_8.
28. **Fit Model 2** (refined OLS model) with 7 statistically significant predictors: CRIM, CHAS, NOX, RM, DIS, PTRATIO, LSTAT.
29. **Evaluate both models** using RMSE, MAE, MAPE, and R-squared on training and test sets.
30. **Select Model 2** as the preferred model based on parsimony, interpretability, test performance, and economic alignment.
31. **Verify four regression assumptions:** residual mean = 0, homoscedasticity (Goldfeld-Quandt test), linearity (residual plot), and normality (Q-Q plot and histogram).
32. **Apply 10-fold cross-validation** — confirm $R^2 = 0.729$ (±0.239) and MSE = 0.040 (±0.023).
33. **Extract and interpret final regression coefficients** — translate coefficients into percentage-based business insights.
34. **Derive actionable insights and business recommendations** across pricing, investment, policy, marketing, and customer segmentation.

35. **Communicate findings to stakeholders** with clear, quantitative rationale grounded in the coefficient estimates and validated model assumptions.